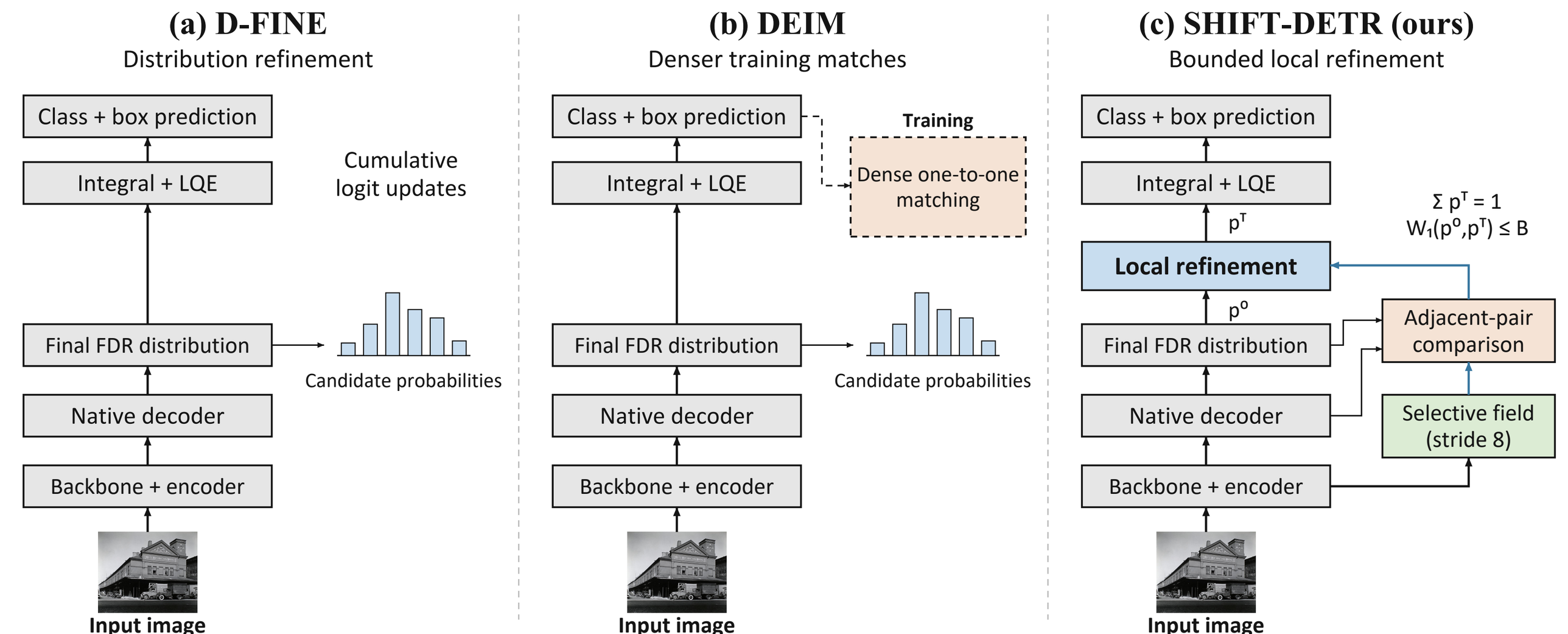

(a) D-FINE
Distribution refinement
Class + box prediction
Integral + LQE
Cumulative logit updates
Final FDR distribution
Candidate probabilities
Native decoder
Backbone + encoder
Input image
(b) DEIM
Denser training matches
Class + box prediction
Training
Dense one-to-one matching
Integral + LQE
Final FDR distribution
Candidate probabilities
Native decoder
Backbone + encoder
Input image
(c) SHIFT-DETR (ours)
Bounded local refinement
Class + box prediction
Integral + LQE
p^T
$\Sigma\, p^T = 1$
$W_1(p^o, p^T) \le B$
Local refinement
p^o
Adjacent-pair comparison
Final FDR distribution
Native decoder
Selective field (stride 8)
Backbone + encoder
Input image